Taha Canturk

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Technical Skills

Languages: C, C++ (C++17–23), Python, Java, C#, Verilog, Bash / POSIX shell
Systems & Software: Linux/Unix/Windows; memory management; multithreading & concurrency; process management; filesystem operations; CLI/GUI development; legacy code maintenance; debugging; test automation
Networking & Communications: TCP/UDP/IP; IPv6; peer-to-peer networking; asynchronous I/O; socket programming; CAN; message serialization/parsing; endianness; industrial equipment communication
Embedded & Firmware: Embedded C/C++; Arduino; Raspberry Pi/Pico; microcontroller programming; PWM control; motor drivers; sensor interfacing; CAN firmware testing; real-time communication; hardware/software integration
Electrical & Hardware: Motor test systems; high-voltage/high-current systems; wiring harnesses; soldering/rework; multimeter use; circuit debugging; H-bridge motor drivers; DC motors; logic gates; encoder wiring; dynamometer setup
Controls & Instrumentation: Industrial motor controllers; ABB ACS800/ACS880 drives; ABZ and sin/cos encoders; FOC; IPM motor characterization; MTPA optimization; torque/speed control; Yokogawa power analyzers; data acquisition; sensor integration
Algorithms & Scientific Computing: Numerical methods; numerical optimization; interpolation/regression; arbitrary-precision arithmetic; matrix transformations; modular arithmetic; bitwise operations; performance optimization; benchmarking; statistical analysis
Cryptography & Security: ECC; ECDH; ECDSA; ECIES; AES; ChaCha20; HMAC; CMAC; HKDF; SHA-256/SHA-512; key exchange; authenticated encryption; cryptographic testing; cryptanalysis
Testing & Performance: Unit testing; regression testing; integration testing; automated testing; benchmarking; profiling; memory debugging; race-condition debugging; performance analysis; large-scale workload testing
Libraries & Frameworks: Boost.Asio; Crypto++; Qt5; PyQt5; PySide6; NumPy; Pandas; Matplotlib; QuTiP
Tools & Development: Git/GitHub/GitLab; GNU Make; Makefiles; CMake; GDB; Valgrind; Wireshark; Docker; Quartus Prime; Android Studio; Jupyter Notebook; LaTeX; Vim
Security Tools: Kali Linux; Metasploit; W4AF; Commix; Wireshark

Relevant Experience

Mechatronics Engineering Assistant – Co-op – Accelerated Systems Incorporated Apr 2026 – Aug 2026

Mechatronics Engineering Assistant – Part-Time – Accelerated Systems Incorporated Sep 2026 – Present

- Commissioned a 200 Nm dynamometer by debugging and extending a 30,000+ line legacy Python codebase and integrating ABB ACS880 drive control, expanding available torque measurement capacity by 4×.
- Independently architected a multithreaded Python/PySide6 test-automation framework to replace hours-long manual motor-testing procedures with reusable automated workflows, integrating hardware interfaces, algorithm execution, watchdog monitoring, data acquisition, and visualization.
- Developing custom MTPA calibration algorithms for IPM motors by evaluating published methods and experimentally validating optimization approaches; generating multidimensional LUTs for FOC-based electric drives, including motors intended for Canadian military vehicles, to address customer demand and support a potential new revenue stream.
- Eliminated a longstanding data-acquisition bottleneck that had limited dynamometer logging to 1 s intervals for years; accelerated motor-controller data access by 20× through P-Table integration and .NET API optimization, alongside Yokogawa processing improvements, enabling 100 ms logging intervals (10× higher logging frequency).
- Integrated ABB encoder modules, ABZ and sinusoidal feedback devices, and Yokogawa instrumentation into high-power motor test systems; performed emergency-stop integration, electrical rework, and hardware-level troubleshooting.

Firmware Developer – Midnight Sun Design Team | *C, CAN, Embedded Systems* 2025

- Developed C-based CAN smoke tests for the WS22 motor controller, validating communication, message handling, and controller responses at the embedded firmware level.
- Implemented CAN message serialization and parsing using C structs and unions, accounting for endianness, field widths, byte ordering, and memory layout when mapping payloads to typed data.
- Built transmit/receive test cases and inspected raw hexadecimal CAN frames to verify payload formatting and identify communication and serialization errors.
- Debugged low-level controller communication by tracing data from C data structures through CAN payloads and comparing transmitted and received byte representations.

Peer Tutor – Embedded Systems & Robotics

- Guided students in wiring and programming servo and stepper motors using Arduino based microcontrollers.

- Helped debug hardware issues including incorrect wiring, voltage/current limitations, and signal timing problems using datasheets and multimeters.
- Supported students in understanding control fundamentals and sensor-based feedback.
- Presented a technical lecture on distance sensors.

Lead Web Developer – MechMania Robotics Competition

- Developed an interactive web application.
- Participated as a speaker at the team's

Automotive Shop Experience

- Diagnosed and repaired mechanical issues on vehicles in a hands-on shop environment.
- Disassembled and reassembled engine components using proper tools and procedures.
- Maintained clean and organized workspace while strictly following safety procedures.
- Completed tasks with minimal supervision.

Construction & Roofing Work

- Transported and loaded heavy materials (50+ lbs) across job sites under time constraints.
- Prepared and mixed cement manually using shovels and on-site tools.
- Sustained consistent work pace during extended periods of heavy lifting and manual labor.
- Followed safety practices in high-risk work environments.

Woodworking Projects

- Designed and built a coffee table and carved wooden cup using workshop machinery.
- Operated saws, drills, and sanding equipment safely and accurately.
- Measured, cut, and assembled materials with strong attention to detail.

Selected Projects

Secure Peer-to-Peer Messaging System | *C++23, Boost.Asio, Crypto++, Qt5, Shell, CMake* 2023 – 2025

- Designed and implemented an asynchronous TCP/IPv6 messaging system using Boost Asio with non-blocking I/O.
- Integrated secure session establishment via ECDH and integrated AES/ChaCha authenticated encryption with HMAC/CMAC/ECDSA verification (Crypto++); built Qt5 GUI for user interaction and session control.
- Implemented unit and regression tests with performance benchmarking; debugged memory and race conditions using Valgrind and GDB.

Big Integer Arithmetic Library | *C++, Makefile* Feb 2023 – Mar 2024

- Designed a template-based fixed-width unsigned integer system with compile time configurable bit sizes using 64-bit segmented storage.
- Implemented full arithmetic, bitwise, comparison, stream, and type-conversion operators with constexpr support.
- Built custom hex/oct string parsing, cross-width casting, and bounded random number generation; verified memory safety using Valgrind.

Blockchain Network Simulator | *Python, Jupyter-Notebook* Dec 2022 – Jul 2023

- Built peer-to-peer blockchain simulator with TCP-based node communication, transaction validation, concurrent mining logic, and wallet key management.
- Implemented block propagation, mempool handling, and proof of work consensus mechanism.
- Designed modular components separating networking, UI, and cryptographic verification.

Embedded Scripting IDE | *Python, PyQt5, LaTeX* Mar 2023 – Jun 2023

- Designed cross-platform desktop IDE with syntax highlighting, file explorer, theming engine, and persistent JSON configuration.
- Implemented Duckyscript parsing and payload management with version history and safe-save tracking.

Elliptic Curve Cryptography Library | *Python, PyPI* Mar 2023 – Jun 2023

- Developed a modular ECC based cryptographic library implementing ECIES encryption and ECDSA signing/verification.
- Integrated HMAC, AES, ECDH key exchange, and HKDF to support authenticated encryption workflows.
- Implemented in compliance with SEC 1 standards and built automated tests validating end-to-end cryptographic pipelines.

Remote Control Arduino Car | *C++, Java, Arduino* Mar 2024 – Apr 2024

- Designed/developed the hardware such as wiring, H-bridge,arduino, DC motors, logic gates, etc.
- Programmed Bluetooth communication to be controlled by an android phone through Android Studio.

SIMD / Performance Optimization Project | *C++, Latex, Python* 2025 – 2026

- Optimized a low-level C++ data-transformation pipeline through cache-aware data layout, vectorization, and parallelization; reduced transformation time from 642 ms to 59 ms per 1M operations ($\sim$11$\times$ speedup).
- Designed endian-aware byte-to-word transformations and benchmarked multiple implementations to evaluate memory-layout and performance tradeoffs.
- Analyzed cache behavior and transformation costs to identify bottlenecks and guide successive optimization iterations.

Cryptographic Stream Cipher Research (C++) | *C++, Latex, Python* 2025 – 2026

- Designed and implemented a symmetric stream-cipher key transformation combining dynamic byte-level permutation with 32-bit ARX operations (modular addition, rotations, XOR) to provide per-message key diffusion and confusion.
- Engineered the C++ implementation around low-level memory representations, explicit endianness handling, cache locality, vectorization, and parallel execution; iterated through multiple designs to reduce transformation overhead by 11$\times$.
- Developed and benchmarked the transformation across concurrency and large-input workloads, including thread-safe encryption-index generation and automated statistical/cryptanalysis tests for avalanche, collision, and differential behavior.

Cryptanalysis CLI Tool | *C++, Python, Make* Oct 2022 – Apr 2024

- Developed multithreaded C++ CLI tool for plaintext recovery in one-time-pad key reuse scenarios.
- Implemented concurrent workload distribution for large ciphertext datasets with memory-safe buffer handling.
- Designed interactive terminal interface with structured error handling and deterministic output formatting.

Linux Systems Programming Experiment | *C++, C, Bash, Linux* 2022 – 2023

- Developed experimental Linux programs exploring daemon processes, process creation, multithreading, filesystem operations, and system resource management.
- Implemented concurrent C++ programs using `std::thread`, process forking, dynamic memory allocation, and Linux system interfaces to study CPU and memory resource behavior.
- Built Bash automation and system-service configurations to experiment with process startup, background execution, filesystem traversal, and persistent process management.
- Used Linux system utilities, POSIX APIs, and low-level process controls to investigate operating-system behavior under sustained resource and process workloads.

Birthday Problem Approximation Analysis | *Python, NumPy, Numerical Methods, LaTeX* Feb 2025 – Apr 2026

- Investigated numerical approximations to the birthday problem by deriving an alternative expression for the number of samples required to reach a 50% collision probability.
- Implemented high-precision numerical experiments using Python `Decimal` arithmetic to compare the derived approximation against the standard birthday-problem approximation across varying numbers of possible outcomes.
- Implemented a secant-method solver with up to 200-digit precision to numerically solve the derived equation and evaluate approximation error.
- Automated 100-iteration comparisons and generated plots of approximation differences, finding the derived expression produced closer results than the standard approximation in approximately 80% of tested cases.

AES Encryption Implementation | *C#, C++, Python* 2022

- Implemented AES-128, AES-192, and AES-256 encryption and decryption in C#, including key expansion, S-box transformations, Galois-field multiplication, and block processing.
- Implemented core AES transformations using byte-level operations, bitwise arithmetic, lookup tables, and fixed-width integer manipulation.
- Developed supporting C++ and Python programs for generating inputs and experimenting with hashing and encryption performance.

Experimental Cryptographic Hash Construction | *C++, Python, NumPy, Jupyter Notebook* 2023

- Experimented with the design and implementation of a custom 512-bit hash construction in C++ and Python using matrix transformations, modular arithmetic, bitwise operations, and elliptic-curve operations.
- Developed an experimental preprocessing and compression pipeline operating on 8$\times$8 matrices with modular arithmetic over a 64-bit prime field and Montgomery-ladder elliptic-curve multiplication.
- Built automated test vectors and diffusion experiments across up to 65,535 generated inputs to examine the construction's output behavior and implementation characteristics.

- Developed Python/NumPy tooling for generating transformation constants and elliptic-curve parameters, and used GDB and Valgrind to debug and validate the implementation.

Grocery List Android App | *Java, Android Studio* 2023

- Software lead for a group Android Studio project developing a grocery list application.
- Developed an Android application for creating and managing multiple grocery lists, including adding, removing, renaming, and checking items.
- Implemented item validation to prevent duplicate items and confirmation dialogs for list and item removal.
- Worked in a group development environment utilizing CI/CD pipelining and project management practices.

BB84 Quantum Key Distribution Simulation | *Python, QuTiP, NumPy* 2022

- Implemented a simulation of the BB84 protocol using the QuTiP and NumPy packages.
- Implemented photon polarization encoding and decoding using computational and diagonal bases to generate shared secrets between Alice and Bob.
- Used the quantumrandom package for generating real random numbers.

Open-Source Contributions

NASA CryptoLib: Fixed memory leaks and logic errors; improved maintainability.
W4AF Security Framework: Resolved build issues and improved code quality.
Completion Tool: Documentation updates & improvements.

Education

University of Waterloo Waterloo, ON
BASc, Computer Engineering, Honours, Co-op *2025 – 2030*

- President's Scholarship of Distinction